

Attendance Management System API

Project Overview

Attendance Management System API is a Flask-based REST API for managing students, subjects, and attendance records.

The project uses Python, Flask, and MySQL.

Features

- Student Management
- Subject Management
- Mark Attendance
- Prevent Duplicate Attendance
- View Attendance by Student
- View Attendance by Date
- Calculate Overall Attendance Percentage
- Calculate Subject-wise Attendance Percentage
- Input Validation and Error Handling

Technologies Used

- Python
- Flask
- MySQL
- REST API
- Postman

Project Modules

1. Student Module
2. Subject Module
3. Attendance Module
4. Percentage Module

Environment Setup

Make sure the following are installed:

- Python 3
- MySQL
- Postman

Step 1: Install Dependencies

Open the project folder in the terminal and run:

```
pip install -r requirements.txt
```

Step 2: Configure Database

Create a MySQL database for the project.

Create a `.env` file and add the required database configuration.

Use `.env.example` as a reference for the required environment variables.

Step 3: Run the Project

Run the Flask application:

```
python app.py
```

The API will start on the local Flask server.

Step 4: Test the API

Open Postman and test the API endpoints.

The project supports:

- Create Subject
- Get Subjects
- Mark Attendance
- Check Duplicate Attendance
- Get Attendance by Student
- Get Attendance by Date
- Get Overall Attendance Percentage
- Get Subject-wise Attendance Percentage

Testing

API tests are available in:

```
test_api.py
```

The tests cover input validation, student creation, attendance creation, duplicate attendance, and attendance percentage calculations.

Project Documentation

The repository contains:

- `statement.md` - Problem Statement
- `architecture.md` - System Architecture
- `workflow.md` - Project Workflow
- `use_case.md` - Use Case Diagram
- `component_diagram.md` - Component Diagram
- `sequence_diagram.md` - Sequence Diagram
- `er_diagram.md` - ER Diagram
- `design_decisions.md` - Design Decisions
- `PROJECT_REPORT.md` - Project Report

Screenshots

Screenshots of the API operations and results are included in the repository.

Future Enhancements

- Low Attendance Alert
- Student Login
- Admin Dashboard
- Attendance Reports

Conclusion

The Attendance Management System API provides a simple way to manage student attendance using Flask REST API and MySQL.